

# Introduction to DATA 606

## Statistics & Probability for Data Analytics

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Fall 2025

# Agenda

- About your instructor
- Syllabus
- Class meetups
- Course Schedule
- Assignments (how you will be graded)
  - Participation
  - Labs
  - Data Project
  - Exams
- Software
  - The DATA606 R Package
  - Using R Markdown

# A little about me...

- Assistant Professor at CUNY in Data Science and Information Systems
- Principal Investigator for a Department of Education Grant to develop and test the Diagnostic Assessment and Achievement of College Skills ([www.DAACS.net](http://www.DAACS.net))
- Authored over a dozen R packages including:
  - [likert](#)
  - [ShinyQDA](#)
  - [DTedit](#)
  - [login](#)
- Specialize in propensity score methods. Three new methods/R packages developed include:
  - [multilevelPSA](#)
  - [TriMatch](#)
  - [PSAboot](#)

# Also a Father...





# Runner...





# And photographer.



# A little about Angela...



**HUNTER**



**UNIVERSITY AT ALBANY**  
State University of New York

**RUTGERS**  
THE STATE UNIVERSITY  
OF NEW JERSEY

**CUNY School of  
Professional Studies** | **CUNY**



Diagnostic Assessment & Achievement of College Skills



# Teaching Experience

- Introduction to Statistics in Social Sciences
- Special Issues in Testing
- Evaluation
- Motivation in Education
- Introduction to the Psychological Processing of Schooling
- Educational Psychology in Adolescent Development



# Homeowner







Syllabus and course materials are here: <https://fall2025.data606.net>

The site is built using [Quarto](#) and hosted on [Github](#). Each page of the site has a "Edit this page" link at the bottom right, use that to start a pull request on Github.

We will use Brightspace primary for submitting assignments only. Please submit a PDF or link to the built HTML (e.g. Rpubs, [Github](#))

PDFs are preferred for the homework as there is some LaTeX formatting in the R markdown files. The `tinytex` R package helps with install LaTeX, but you can also install LaTeX using [MiKTeX](#) (for Windows) and [BasicTeX](#) (for Mac) See this page for more information:

<https://fall2025.data606.net/course-overview/software/>



# Meetups

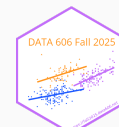
We will have meetups on Wednesday evenings at 8:00pm.

Meetups will be recorded and made available the next day on the [course website](#).

Though attending live is not strictly required, **I expect everyone to watch the lectures during the week.** I use the class meetups to convey important information and announcements. Very often I will cover some topics not in the textbook. Students who attend the meetups tend to do well on the assignments.

**One Minute Papers** - Complete the one minute paper after each Meetup (whether you watch live or watch the recordings). It should take approximately one to two minutes to complete. This allows me to 1) verify you have attended/watch the meetup and 2) get feedback about what you learned and what you may still be unclear.

**Please note:** *Students who participate in this class with their camera on or use a profile image are agreeing to have their video or image recorded solely for the purpose of creating a record for students enrolled in the class to refer to, including those enrolled students who are unable to attend live. If you are unwilling to consent to have your profile or video image recorded, be sure to keep your camera off and do not use a profile image. Likewise, students who un-mute during class and participate orally are agreeing to have their voices recorded. If you are not willing to consent to have your voice recorded during class, you will need to keep your mute button activated and communicate exclusively using the "chat" feature, which allows students to type questions and comments live. [Click here for CUNY's camera use policy](#)*

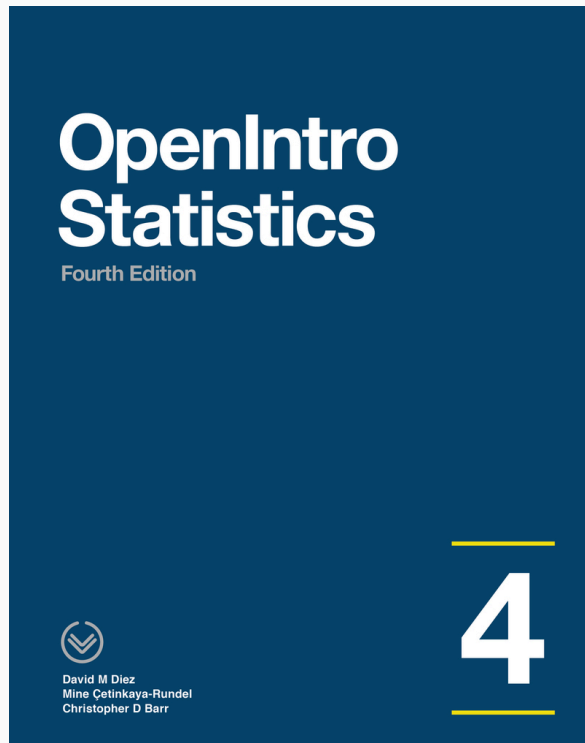


# Schedule

| Start                      | End                        | Topic                                      |
|----------------------------|----------------------------|--------------------------------------------|
| Monday, August 25, 2025    | Sunday, August 31, 2025    | Chapter 1 - Intro to Data, R, and RStudio  |
| Monday, September 01, 2025 | Sunday, September 14, 2025 | Chapter 2 - Summarizing Data               |
| Monday, September 15, 2025 | Sunday, September 21, 2025 | Chapter 3 - Probability                    |
| Monday, September 22, 2025 | Sunday, September 28, 2025 | Chapter 4 - Distributions                  |
| Monday, September 29, 2025 | Sunday, October 05, 2025   | Chapter 5 - Foundation for Inference       |
| Monday, October 06, 2025   | Sunday, October 12, 2025   | Chapter 6 - Inference for Categorical Data |
| Monday, October 13, 2025   | Sunday, October 19, 2025   | Chapter 7 - Inference for Numerical Data   |
| Monday, October 20, 2025   | Sunday, November 02, 2025  | Chapter 8 - Linear Regression              |
| Monday, November 03, 2025  | Sunday, November 09, 2025  | Chapter 9 - Logistic Regression            |
| Monday, November 24, 2025  | Sunday, November 30, 2025  | Thanksgiving                               |
| Monday, December 01, 2025  | Sunday, December 07, 2025  | Intro to Bayesian Analysis                 |
| Monday, December 08, 2025  | Sunday, December 14, 2025  | Final Exam                                 |

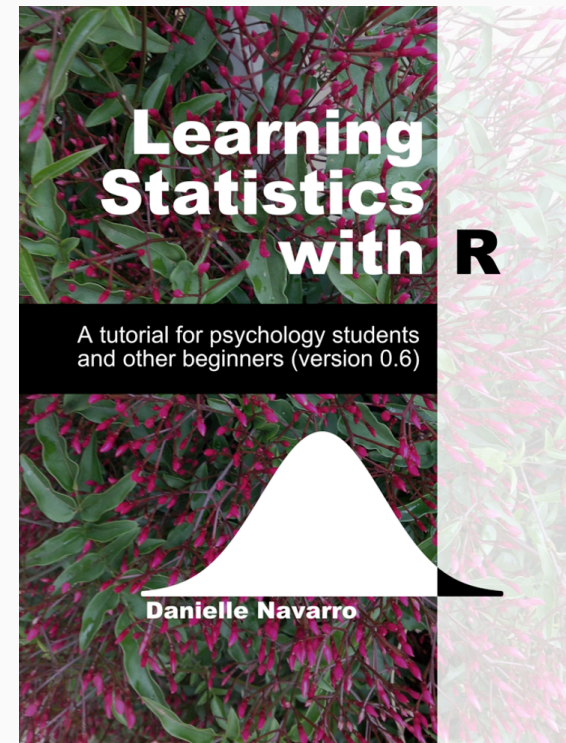
Diez, D.M., Barr, C.D., & Çetinkaya-Rundel, M. (2019). *OpenIntro Statistics (4th Ed)*.

This will be our primary textbook for most of the semesters. Our goal is to cover all the chapters.



Navarro, D. (2018, version 0.6). *Learning Statistics with R*

This textbooks has a chapter on Bayesian analysis that we will use at the end of the semester.





# Assignments

- Participation (10%)
  - DAACS
  - One Minute Papers
- Labs (35%)
  - Labs are designed to introduce to you doing statistics with R.
  - Answer the questions in the main text as well as the "On Your Own" section.
- Data Project (30%)
  - This allows you to analyze a dataset of your choosing. Projects will be shared with the class. This provides an opportunity for everyone to see different approaches to analyzing different datasets.
- Exams
  - Midterm (10%)
  - Final exam (15%)

# Use of Artificial Intelligence (AI)

First, AI is a marketing term. I prefer to be more specific regarding what we are doing:

1. Machine Learning (ML) - This course, along with IS382, will provide the foundations for how ML algorithms work. Generally speaking, the goal is to predict some known (and sometimes unknown in the case of unsupervised learning models) outcome.
2. Large Language Models (LLM) - This is often what people mean when they say AI. This includes products like ChatGPT, Anthropic, Google Gemini, etc. LLMs generate text, images, videos, etc. from a prompt.

The goal of this course is for *you to develop the foundation knowledge and skills to do statistics*. Using chat bots to do the assignments subverts this goal. **The content generated by LLMs is often wrong!** If you use LLMs to assist in completing the assignments, **you must include the prompt and response in your submission.**

# Communication

- Slack Channel: <https://cuny-msds.slack.com>
  - [Click here to join the group](#)
- Email: [jason.bryer@cuny.edu](mailto:jason.bryer@cuny.edu)
- Phone/Zoom: Please email to schedule a time to meet.
- Office hours by appointment.



This is an applied statistics course so we will make extensive use of the **R statistical programming language**.

Install **R** and **RStudio** on your own computer. I encourage everyone to do this at some point by the end of the semester. I have instructions on the course website here:

<https://fall2025.data606.net/course-overview/software/>

You will also need to have **LaTeX** installed as well in order to create PDFs. The **tinytex** R package helps with this process:

```
install.packages('tinytex')
tinytex::install_tinytex()
```

The **DATA606** R package contains many data sets and functions we will use throughout the semester. It also has a `startLab` function that will copy each of the labs to your current working directory. Use the following commands to install the package (only necessary once per R installation):

```
remotes::install_github('jbryer/DATA606')
```

To start the first lab...

```
DATA606::startLab('Lab1')
```

This will copy the R markdown file and any supporting files to your current working directory. Use the "Knit" button in R Studio to build a PDF of the document.

# Next steps...



Before Wednesday (August 31st):

- Complete this Google form: <https://forms.gle/6RmyyWJ97L7iJB9>
- Go to <https://cuny.daacs.net> and complete the self-regulated learning assessment
- [Join the Slack channel](#)

Then:

- Start Lab 1 (due August 31st)



# Good luck with the semester!

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